

 HF3-SW-22 HACKATHON PROJECT

TRUTHLAYER

The Trust Layer for AI

Claim-level factual verification & evidence routing for LLM-generated information

 Raw AI Output →  TRUTHLAYER →  Evidence Verification →  Verified Knowledge

Team: VeriMind AI

Hackathon: Tech Vision Hackathon 2026

Track: AI Safety & Trust Infrastructure

AI Can Sound Right Even When It Is Wrong

01 // THE PROBLEM

LLMs generate answers that are **fluent, confident, and persuasive**—making hallucinations exceptionally hard to catch.

💬 Confident AI Answer



👤 Looks Trustworthy & Fluent



⚠️ Individual Claims are Factually Incorrect

Core Challenge: The problem is not just whether an AI answer is wrong—it is knowing *WHICH* specific parts are wrong and *WHY*.

Standard LLM Response

UNVERIFIED OUTPUT

CLAIM 1: "India became independent in 1947."

CORRECT

CLAIM 2: "India has 36 states."

INCORRECT

CLAIM 3: "India's GDP is \$10 trillion."

NEEDS VERIFICATION

The Citation Problem in Current AI

02 // TRUST ISSUES

❌ Why Simple Citations Fail

Attaching a URL at the end of an AI response does not mean every claim inside the response is supported by that source.

AI Answer (Contains 5 Factual Claims)



1 Generic Citation Attached

❓ Does source support Claim #3?

UNKNOWN TO USER

Factuality

"Is the claim actually true in the real world based on empirical evidence?"

Faithfulness

"Is the claim directly supported by the specific context/evidence provided?"

Confabulation

AI generating plausible-sounding but completely unsupported or incorrect details with high confidence.

Key Insight: Users need evidence grounded at the **CLAIM level**, not just citations at the answer level.

Where AI Hallucinations Matter Most

03 // HIGH STAKES

Healthcare

Incorrect AI medical guidance can influence critical health decisions and dosage errors.

Legal

Fabricated court case citations or unsupported claims misinform legal briefs.

Education

Students internalize unverified historical or scientific facts presented as truth.

Finance

Incorrect financial statistics or revenue metrics lead to flawed investment choices.

Research

Ghost references and non-existent papers waste valuable research hours.

Software Dev

Outdated or hallucinated API parameters cause security vulnerabilities and broken builds.

As AI becomes embedded in high-stakes decision-making, **transparent evidence verification is mandatory.**

Introducing TruthLayer

TruthLayer is an AI-powered verification layer that evaluates LLM responses at the granular claim level.

- 01 Extract Atomic Claims from AI Text
- 02 Retrieve Multi-Source Empirical Evidence
- 03 Evaluate Source Credibility & Agreement
- 04 Calculate Confidence & Composite Trust Score

Transparent 3-Tier Classification

VERIFIED



Directly supported by authoritative evidence.

FALSE



Contradicted by retrieved facts and sources.

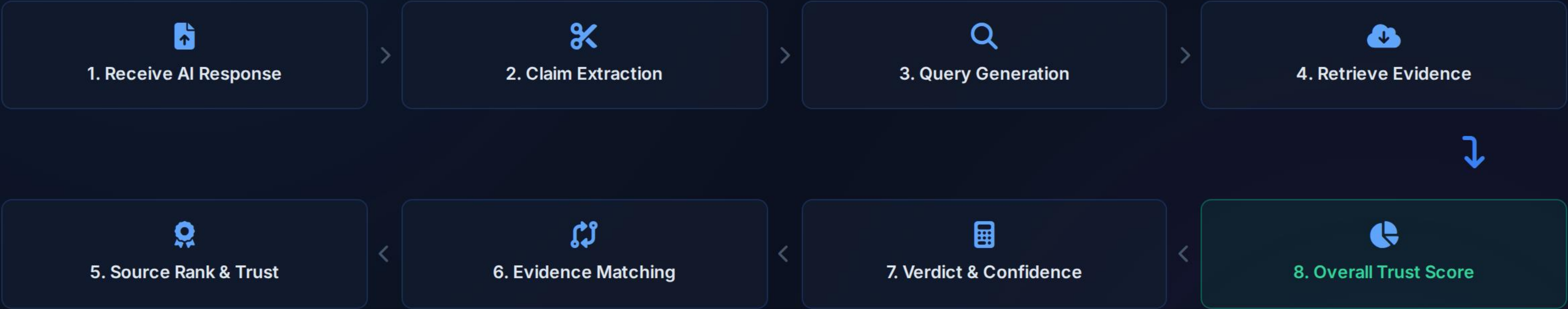
UNVERIFIABLE



Insufficient or conflicting web data available.

Crucial: TruthLayer never forces uncertain information into binary true/false.

How TruthLayer Works: 8-Step Pipeline

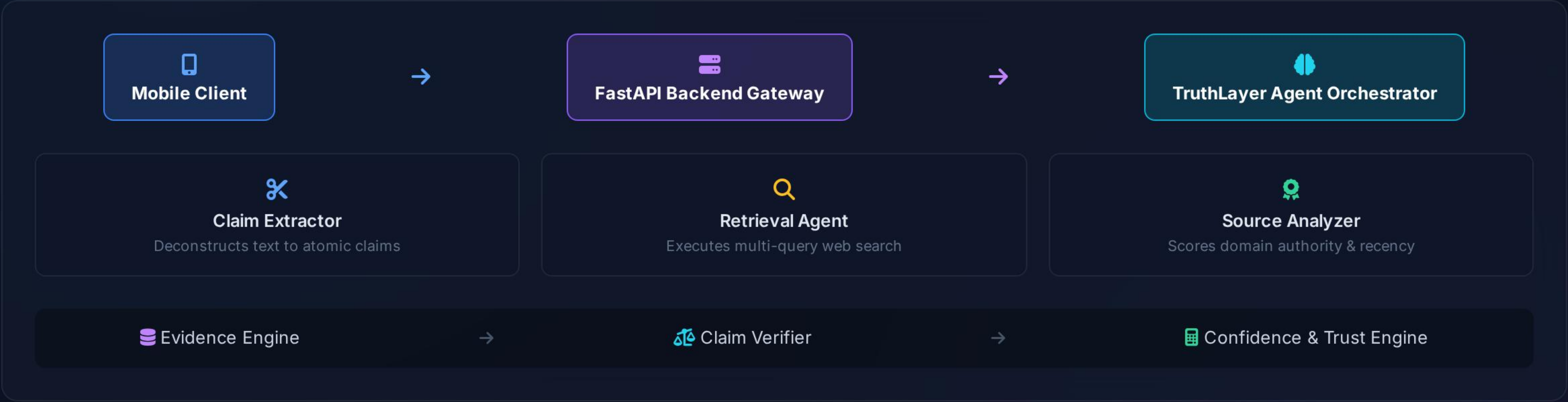


⚡ Automated Querying

🛡️ No Self-Verification Bias

⚙️ Dynamic Confidence Weighting

Hallucination Hunter Agent Architecture



Modular Loop: The Agent autonomously triggers secondary retrieval rounds if initial evidence confidence is insufficient.

Claim-Level Verification in Action

INPUT AI ANSWER:

"India became independent in 1947. India has 36 states. India's GDP is \$10 trillion."

VERIFIED	Claim 1: "India became independent in 1947."	98%
Evidence: Indian Independence Act 1947 (Official Archives)		Confidence
FALSE / CONTRADICTED	Claim 2: "India has 36 states."	94%
Evidence: Govt Portal confirms 28 States and 8 Union Territories.		Confidence
UNVERIFIABLE / OUTDATED	Claim 3: "India's GDP is \$10 trillion."	52%
Evidence: World Bank 2024 lists GDP ~\$3.75 Trillion. Projected target for future.		Confidence

 Source Evaluation Matrix

Domain Authority	.gov / .edu / Verified APIs
Semantic Relevance	Vector Cosine Similarity
Temporal Recency	Timestamp Decay Factor
Cross-Source Agreement	Independent Consensus

SAMPLE SOURCE AUDIT

SCORE: 92 / 100

- ✔ Government Official Portal (Primary Source)
- ✔ High Direct Textual Entailment (94%)
- ✔ Updated within last 30 days

Golden Rule: The verifier relies on actual retrieved evidence rather than simply prompting an LLM to self-reflect.

Trust Score & Explainable Verdicts



8 Verified 2 False 2 Unverifiable

Detailed Claim Audit

CLAIM: "India has 36 states."

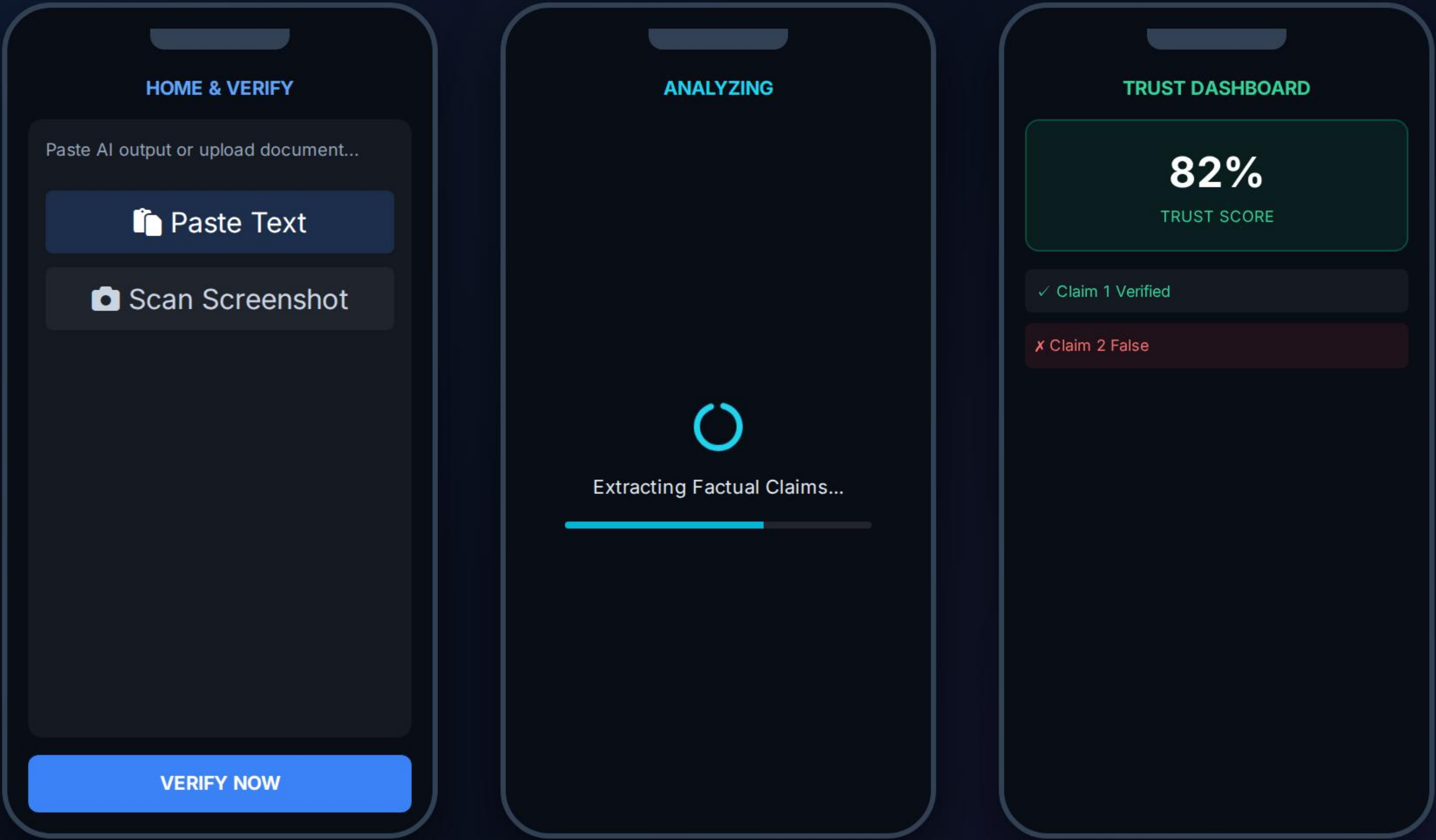
VERDICT: FALSE

CONFIDENCE: 94%

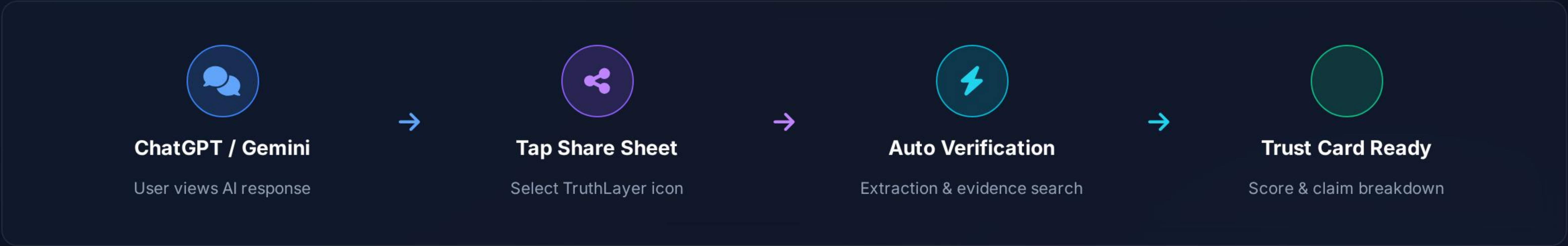
Why? Retrieved official government database explicitly confirms 28 states and 8 Union Territories, directly contradicting the claim.

Retrieved Sources: india.gov.in (Official), PIB Census Archive

Mobile App Experience: Built for Speed



Share-to-Verify: Seamless Mobile Flow



✂️ Eliminates manual copy-pasting—verify AI claims natively right inside your daily workflow.

Beyond Basic Fact Checking

12 // CAPABILITIES

Input Modes

- Share-to-Verify Integration
- Direct Text Clipboard Paste
- Screenshot OCR Extraction
- PDF / DOCX File Parsing

Verification

- Claim Deconstruction
- Dynamic Query Generation
- Source Credibility Scoring
- Contradiction Detection

Intelligence

- Self-Correction Engine
- Verified Answer Rewrite
- Iterative Re-verification
- Multi-Model Agreement

UX & Control

- Verification Audit History
- Search & Tag Filters
- PDF Exportable Audit Reports
- Zero-Data Retention Mode

Self-Correction: From Detection to Truth

13 // CORRECTION

1. DETECTED HALLUCINATION

"India has 36 states."

↓ Retrieve Evidence & Context

2. GENERATE FACTUAL CORRECTION

"India currently has 28 states and 8 Union Territories."

↓ Re-verify Corrected Output

3. VERIFIED REWRITTEN ANSWER

100% Factually Grounded Response Ready

Integrity Guarantee

The self-correction module **never silently overwrites** the original AI response.

Users always see a clear diff comparison showing the original hallucination alongside the evidence-backed correction.

Built With: Production Tech Stack



Mobile Frontend

React Native (Expo)
TypeScript
Tailwind React Native



Backend Services

Python 3.11
FastAPI
PostgreSQL & Redis



AI & Orchestration

LangGraph / Agentic Loops
OpenAI GPT-4o / Claude 3.5
Custom Claim Parsers

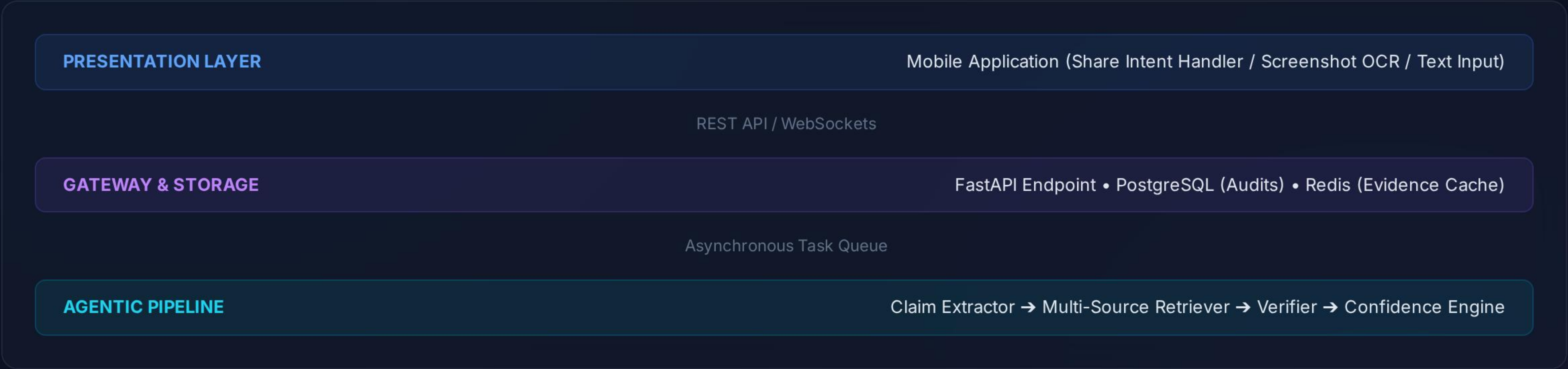


Retrieval & Vector

Tavily & Serper Web APIs
Qdrant Vector DB
SentenceTransformers

FULLY IMPLEMENTED BACKEND & MOBILE CORE

System Architecture Overview



Trust Starts With Privacy

16 // SECURITY

Data & Key Protection

- ✓ API keys strictly isolated on secure backend servers.
- ✓ TLS 1.3 encrypted data transmission in transit.
- ✓ User-controlled verification history deletion.
- ✓ Optional "Do Not Store My Responses" toggle.

Truth Ethics Policy

- ✓ Zero Synthetic Evidence: No LLM-generated fake citations.
- ✓ Uncertain claims remain flagged as Unverifiable.
- ✓ Strict attribution to original verified URLs.

Evaluation Dimensions

- 1. Claim Extraction:** Precision of breaking text into atomic statements.
- 2. Retrieval Quality:** Relevance of retrieved web documents.
- 3. Classification:** Correct classification (Verified vs False vs Unverifiable).
- 4. Confidence Calibration:** Alignment of score with empirical ground truth.

Target Metrics Benchmark

Precision / Recall	F1 Score
False Positive Rate	End-to-End Latency

* Frame based on standard evaluation protocol for factual grounding in language models.

Evaluation Strategy: Human vs TruthLayer



Why Not Just Use Basic RAG?

APPROACH	RETRIEVES EVIDENCE	CLAIM-LEVEL VERIFICATION	CONFIDENCE SCORE	EXPLAINS VERDICT
Standard LLM	✗ No	✗ No	— Weak	✗ No
Basic RAG System	✓ Yes	— Limited	— Limited	— Limited
Search Engine Citations	✓ Yes	✗ No	✗ No	— Partial
 TruthLayer	✓ Yes	✓ Granular	✓ Calibrated	✓ Full Audit

Key Differentiator: TruthLayer does not just generate an answer—it rigorously inspects and verifies every claim *inside* the answer.

Future Vision: AI Trust Infrastructure

20 // ROADMAP



Browser Extension

Highlight & verify any text live while browsing ChatGPT, Gemini, or Web pages.



Middleware API

API for AI agents to self-verify before presenting output to end users.



Enterprise Audit

Automated compliance and hallucination risk monitoring for enterprise LLMs.

Why This Matters: Broader Impact

21 // MISSION

“We don’t want people to stop using AI. We want them to **use AI responsibly.**”

Transparency

Demystify AI confidence and highlight empirical sources.

Informed Usage

Empower users to spot hallucinations instantly.

Velocity

Verify complex information in seconds via mobile share.

 SWITCH TO LIVE DEMO



Conclusion & Summary

23 // SUMMARY

- ✓ Whole-answer scores are insufficient—TruthLayer verifies claim by claim.
- ✓ Retrieved multi-source evidence ensures un-biased factual grounding.
- ✓ Share-to-Verify brings claim auditing natively to mobile users.

**“Don’t Blindly Trust AI.
Verify It with TruthLayer.”**



Thank You

TruthLayer — The Trust Layer for AI

 github.com/verimind/truthlayer

 truthlayer.ai

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